

SADStream: Adaptive Semantic Stream Clustering for Dynamic Topic Detection and Tracking

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Abstract

Social-media streams contain continuously arriving, short, noisy, and rapidly changing text. Traditional clustering methods such as K-Means and hierarchical clustering are poorly suited to this setting because they generally assume a finite dataset and, in the case of K-Means, a predefined number of clusters. Stream-clustering methods such as DenStream address evolving data through incremental micro-clusters and temporal fading, but they were not designed specifically for semantic short-text streams.

This research proposes **SADStream**, an online topic detection and tracking framework combining sentence-level semantic representations, density-based micro-clustering, adaptive forgetting, and burst-aware event detection. The system does not require the number of topics to be specified in advance. Each topic maintains its own decay rate according to recent activity, allowing inactive topics to disappear faster while active topics remain available for tracking. A compact long-term memory stores representations of expired topics so that recurring topics can be distinguished from genuinely new ones.

The method will be evaluated on social-media event-detection datasets such as Event2012 and Event2018. Evaluation will measure clustering quality, event-detection accuracy, detection delay, processing throughput, and memory usage. The main research question is whether adaptive semantic stream clustering can improve dynamic event tracking while retaining bounded-memory online operation.

1 Introduction

Social-media platforms generate unbounded streams of short textual messages. Real-world events such as natural disasters, transportation incidents, sporting events, and breaking news may cause new topics to appear suddenly, evolve rapidly, and disappear shortly afterward.

Let the incoming stream be

$$S = \{(x_1, t_1), (x_2, t_2), \dots\}, \quad (1)$$

where x_i denotes a social-media post arriving at time t_i .

The active topic structure changes over time:

$$C_t \neq C_{t+\Delta}. \quad (2)$$

A topic may be created, grow, drift semantically, disappear, or later recur. Therefore, social-media topic detection is a non-stationary clustering problem.

Traditional clustering methods are inadequate for three main reasons:

1. the complete dataset is never available;
2. the number of topics is unknown and changes over time;
3. older observations should gradually lose importance as the stream evolves.

Existing stream-clustering algorithms such as DenStream address incremental clustering and temporal fading, while short-text methods such as MStreamF introduce explicit forgetting. Modern sentence embeddings improve semantic similarity for short text, but semantic representation, adaptive forgetting, recurring-topic recognition, and breaking-event detection are rarely handled within one online framework.

The main research question is therefore:

Can semantic online clustering with adaptive forgetting improve real-time topic detection and tracking under evolving social-media streams?

SADStream addresses four main responsibilities:

1. semantic representation of incoming posts;
2. incremental topic clustering;
3. temporal adaptation and forgetting;
4. event detection and tracking.

2 Related Work and Research Gap

Stream-clustering algorithms maintain compact summaries instead of storing an entire data stream. DenStream uses potential and outlier micro-clusters together with temporal fading, allowing clusters to appear and disappear without requiring the number of clusters in advance.

Short-text stream clustering has been addressed by methods such as MStream and MStreamF. MStreamF is particularly relevant because it includes a forgetting mechanism for removing outdated clusters. However, these approaches primarily rely on lexical or probabilistic text representations.

Modern topic-modeling systems increasingly use contextual embeddings. BERTopic, for example, clusters transformer-based document embeddings and produces interpretable topic representations. Recent event-detection studies also demonstrate that sentence embeddings can be combined with incremental clustering.

However, three limitations remain:

1. semantic event-detection approaches are often batch- or window-based rather than true bounded-memory stream-clustering systems;
2. traditional stream-clustering algorithms commonly use a fixed global forgetting rate,

$$f(\Delta t) = e^{-\lambda \Delta t}, \quad (3)$$

which assumes all topics have similar lifetimes;

3. completely deleting inactive clusters makes it difficult to distinguish a genuinely new topic from a previously observed topic that has returned.

The proposed research addresses these limitations through

Semantic Representation + Online Micro-Clustering + Adaptive Forgetting + Topic Memory
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3 Proposed Method

The SADStream pipeline is

$$\text{Social Stream} \rightarrow \text{Semantic Representation} \rightarrow \text{Micro-Clustering} \rightarrow \text{Temporal Adaptation} \rightarrow \text{Event Detection.} \quad (5)$$

3.1 Semantic Representation

Each incoming post x_t is converted into a dense semantic embedding:

$$e_t = \text{Encoder}(x_t), \quad (6)$$

followed by normalization:

$$\hat{e}_t = \frac{e_t}{\|e_t\|_2}. \quad (7)$$

Because event-specific information such as hashtags and named entities may remain important, a hybrid similarity function can be used:

$$S(x, C_k) = \alpha S_{\text{semantic}} + \beta S_{\text{lexical}} + \gamma S_{\text{entity}} + \delta S_{\text{hashtag}}. \quad (8)$$

The contribution of these additional signals will be measured through ablation experiments.

3.2 Online Topic Clustering

The system maintains a set of micro-clusters:

$$M_t = \{MC_1, \dots, MC_m\}. \quad (9)$$

Each micro-cluster stores compact statistics:

$$MC_k = (\mu_k, W_k, N_k, T_k), \quad (10)$$

where μ_k is the semantic centroid, W_k is temporal weight, N_k is the number of assigned posts, and T_k contains temporal metadata.

For a new post, the closest compatible cluster is selected:

$$MC^* = \arg \max_k S(e_t, MC_k). \quad (11)$$

If similarity exceeds a threshold, the new observation is absorbed. Otherwise, a new candidate micro-cluster is created.

The cluster weight is updated as

$$W'_k = W_k f_k(\Delta t) + 1, \quad (12)$$

and its centroid becomes

$$\mu'_k = \frac{W_k f_k(\Delta t) \mu_k + e_t}{W'_k}. \quad (13)$$

A candidate cluster is promoted to an active topic when

$$W_k > \theta_p. \quad (14)$$

Thus, the number of active topics can change dynamically.

3.3 Topic Formation and Labeling

Each active micro-cluster represents a candidate topic. Posts are assigned to the most semantically compatible micro-cluster according to the hybrid similarity function defined above. If the highest similarity exceeds the assignment threshold, the post is incorporated into that cluster; otherwise, a new candidate micro-cluster is created.

To improve topic coherence, closely related micro-clusters may also be grouped into a higher-level topic when their semantic centroids and auxiliary features are sufficiently similar. For two clusters MC_i and MC_j , a topic-level similarity score may be defined as

$$S_{\text{topic}}(MC_i, MC_j) = \alpha_c \cos(\mu_i, \mu_j) + \beta_c J(K_i, K_j) + \gamma_c J(E_i, E_j) + \delta_c J(H_i, H_j), \quad (15)$$

where μ_i and μ_j are semantic centroids, K denotes representative keywords, E denotes named entities, H denotes hashtags, and $J(\cdot, \cdot)$ is Jaccard similarity. Clusters whose similarity exceeds a merging threshold θ_M may be treated as belonging to the same topic. This optional aggregation allows several local micro-clusters to represent different semantic variations of the same real-world event while preserving incremental processing.

Each detected topic is assigned both a persistent identifier and a human-readable label. The persistent identifier, such as **Topic-017**, is used internally for tracking the topic through its birth, active, decaying, dead, and recurring states. The human-readable label summarizes the semantic content of the topic.

Topic labels are generated from information accumulated within the cluster. For each topic C_k , the system maintains

$$R_k = (K_k, E_k, H_k), \quad (16)$$

where K_k contains representative keywords, E_k contains salient named entities, and H_k contains frequent hashtags. Representative keywords can be extracted from recent posts assigned to the cluster using class-based TF-IDF, TF-IDF, or another lightweight term-weighting method. Named entities and hashtags provide additional event-specific information that may be more informative than generic high-frequency words.

A topic label is then constructed from the highest-ranked elements of R_k . For example, a cluster containing posts about an earthquake near Tokyo might obtain the internal identifier

$$\text{Topic-017} \quad (17)$$

and the descriptive label

$$\text{"Tokyo earthquake"}. \quad (18)$$

Labels are updated incrementally as additional posts enter the cluster. To avoid unstable topic names, a label is changed only when newly dominant keywords or entities persist for a minimum number of observations or time intervals. The persistent topic identifier remains unchanged even when the descriptive label evolves.

When an inactive topic later reappears, similarity with its archived centroid, keywords, and entities is used to recover its previous topic identity where appropriate. The recurring cluster can therefore retain the same persistent topic identifier while its descriptive label is refreshed from the newly arriving posts.

This separation between *cluster identity* and *topic label* allows SADStream to distinguish computational tracking from human interpretation:

$$\boxed{\text{Post} \rightarrow \text{Micro-Cluster} \rightarrow \text{Topic ID} \rightarrow \text{Human-Readable Label}} \quad (19)$$

3.4 Temporal Adaptation and Forgetting

The primary methodological contribution is cluster-specific adaptive forgetting.

Conventional temporal fading uses a single decay parameter:

$$f(\Delta t) = e^{-\lambda \Delta t}. \quad (20)$$

SADStream instead assigns each topic its own decay rate:

$$f_k(\Delta t) = e^{-\lambda_k(t) \Delta t}. \quad (21)$$

A preliminary formulation is

$$\lambda_k(t) = \frac{\lambda_0}{1 + \eta B_k(t) + \rho A_k(t)}, \quad (22)$$

where $A_k(t)$ represents recent activity and $B_k(t)$ represents burstiness.

Active topics therefore decay more slowly, while inactive topics disappear more quickly.

When

$$W_k < \theta_{\text{inactive}}, \quad (23)$$

the topic is removed from active memory.

Instead of deleting it completely, the system stores a compressed prototype:

$$L_k = (\mu_k, K_k, E_k, t_{\text{start}}, t_{\text{end}}), \quad (24)$$

where K_k and E_k denote representative keywords and entities.

The overall memory therefore consists of

$$\text{Active Memory} + \text{Compressed Topic Archive}. \quad (25)$$

When a new candidate appears, it is compared with archived prototypes. If

$$\text{sim}(C_{\text{new}}, L_j) > \theta_R, \quad (26)$$

the cluster is classified as a recurring topic rather than a completely new topic.

3.5 Event Detection and Tracking

Cluster size alone does not determine whether a topic is breaking. SADStream therefore measures changes in recent activity.

For topic k :

$$A_k(t) = \sum_{x_i \in C_k} e^{-\lambda_k(t-t_i)}. \quad (27)$$

Burstiness is estimated as

$$B_k(t) = \frac{A_k(t) - \mu_{A,k}}{\sigma_{A,k} + \epsilon}. \quad (28)$$

The breaking-event score is

$$\text{Score}_k = w_1 B_k + w_2 N_k + w_3 D_k + w_4 U_k, \quad (29)$$

where B_k is burstiness, N_k is novelty, D_k is semantic density, and U_k is user or source diversity.

An alert is generated when

$$Score_k > \theta_B. \quad (30)$$

Each topic receives a persistent identifier and can occupy one of the following states:

$$\{\text{birth, active, decaying, dead, recurring}\}. \quad (31)$$

4 Research Questions

The study focuses on three primary research questions:

- RQ1.** Does semantic representation improve incremental event clustering compared with lexical representations such as TF-IDF?
- RQ2.** Does cluster-specific adaptive forgetting improve dynamic topic tracking compared with fixed exponential decay?
- RQ3.** Does retaining compressed representations of inactive topics improve recurring-topic recognition without substantially increasing memory usage?

A secondary research question examines whether burst-aware ranking detects emerging events earlier than ranking clusters using message volume alone.

5 Experimental Evaluation

5.1 Datasets

The primary benchmark datasets are:

- Event2012;
- Event2018.

Posts will be processed in chronological order to preserve the streaming setting.

A Vietnamese social-media dataset may be added as an additional language-transfer experiment if suitable data and annotation resources are available.

5.2 Baselines

Table 1 summarizes the minimum baseline set.

Table 1: Proposed experimental baselines.

Model	Purpose
TF-IDF + incremental/window clustering	Lexical baseline
Sentence embedding + incremental clustering	Semantic baseline
DenStream	General stream-clustering baseline
MStreamF	Short-text forgetting baseline
SADStream with fixed decay	Isolates adaptive forgetting
Full SADStream	Proposed model

The most important controlled comparison is

$$SADStream_{\text{fixed}} \text{ vs. } SADStream_{\text{adaptive}}. \quad (32)$$

5.3 Evaluation Metrics

Clustering quality will be measured using

$$ARI, \quad AMI/NMI, \quad B^3F_1. \quad (33)$$

Event-detection performance will use

$$Precision, \quad Recall, \quad F_1. \quad (34)$$

Detection delay for event E is defined as

$$Delay(E) = t_{\text{detected}}(E) - t_{\text{start}}(E). \quad (35)$$

Streaming performance will additionally measure:

- posts processed per second;
- per-post processing latency;
- peak memory consumption;
- number of active micro-clusters.

5.4 Ablation Study

The minimum ablation study compares

$$SADStream_{\text{full}} \quad (36)$$

against

$$- \text{Adaptive Forgetting}, \quad (37)$$

$$- \text{Long-Term Memory}, \quad (38)$$

$$- \text{Semantic Representation}. \quad (39)$$

These experiments determine whether each claimed component contributes materially to the final performance.

6 Expected Contributions

The study targets three primary contributions.

6.1 Semantic Stream-Clustering Framework

A bounded-memory online clustering framework using contextual semantic representations without requiring the number of topics in advance.

6.2 Adaptive Topic Forgetting

A topic-specific decay mechanism

$$\lambda_k(t) \quad (40)$$

that allows clusters with different temporal lifetimes to be forgotten at different rates.

6.3 Recurring-Topic Memory

A compact semantic archive that allows the system to distinguish between

$$\text{New Topic} \tag{41}$$

and

$$\text{Recurring Topic.} \tag{42}$$

The central hypothesis is:

Adaptive semantic stream clustering can track emerging and disappearing social-media topics more effectively than fixed-decay streaming methods while retaining bounded-memory operation.

7 Minimum Implementation Scope

The minimum implementation is restricted to

Sentence Encoder + DenStream-style Micro-Clusters + Adaptive Decay + Topic Archive + Burst Score
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The study does not require:

- training a language model from scratch;
- supervised event classification;
- misinformation detection;
- sentiment analysis;
- full social-network graph modeling;
- sophisticated merge/split detection;
- production-scale distributed deployment.

The minimally successful result is a chronological stream experiment demonstrating whether

$$\text{Adaptive Decay} > \text{Fixed Decay} \tag{44}$$

in topic clustering or tracking quality while maintaining acceptable latency and memory usage.

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